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Applied machine learning to predict 1-year major adverse cardiovascular events in elderly patients after percutaneous coronary intervention

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Abstract

Background Cardiovascular diseases remain the leading cause of mortality worldwide, with elderly patients experiencing the worst prognosis following ST-elevation myocardial infarction (STEMI). Traditional risk stratification models demonstrate suboptimal performance in geriatric patients due to complex risk profiles involving frailty and multiple comorbidities.

Objectives To develop machine learning-based predictive models for one-year major adverse cardiovascular events (MACE) in elderly patients (≥ 65 years) undergoing percutaneous coronary intervention (PCI) for STEMI.

Methods This retrospective cohort study analyzed 1,358 elderly patients who underwent PCI between 2015 and 2021. MACE included cardiovascular death, myocardial infarction, stroke, and revascularization within one year. Eight machine learning algorithms were evaluated: XGradient Boosting (XGB), Random Forest (RF), Logistic Regression, Neural Networks, Support Vector Machines, K-Nearest Neighbors, Decision Trees, and Naive Bayes. The Synthetic Minority Oversampling Technique (SMOTE) was applied to address class imbalance. Model performance was evaluated using various metrics, and SHAP (Shapley Additive Explanations) was used to enhance model interpretability and support clinical decision-making by identifying key risk factors driving predictions.

Results Among the patients (mean age 74.1 ± 6.7 years, 31.8% female), 152 (11.2%) experienced MACE within one year. XGB and RF emerged as the most robust models, achieving area under the receiver operating characteristic curve (AUC) values of 94% and 95%, respectively, with RF demonstrating higher sensitivity (79%) and specificity (96%). SHAP analysis revealed pre-PCI ejection fraction, age, creatinine levels, fasting blood sugar, BMI, and LDL/HDL ratio as the most influential predictors.

Conclusion Machine learning models demonstrated strong predictive performance in predicting one-year MACE in elderly post-PCI patients. By combining high-performance models with SHAP-based explanations, this approach supports transparent, clinically actionable risk stratification and personalized care.

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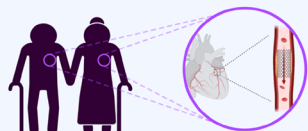
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Keywords Machine learning, Major adverse cardiovascular events, Percutaneous coronary intervention, Myocardial infarction, Elderly

Graphical Abstract

Applied Machine Learning to Predict 1-Year Major Adverse Cardiovascular Events in the Elderly After Percutaneous Coronary Intervention

Study Population and Design

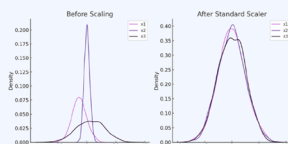


This retrospective cohort study analyzed data from 1,358 patients aged ≥ 65 who underwent PCI

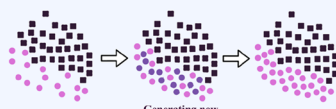
Data Preparation



Feature extraction with identifying significant variables between MACE and Non-MACE groups, followed by elimination of highly correlated variables

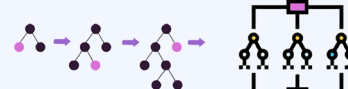


Data was normalized with StandardScaler



Imbalanced data addressed with SMOTE technique

Models Performance



XGradient Boosting

Random Forest

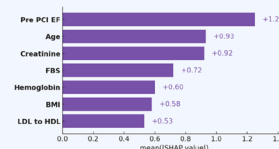
XGB and RF demonstrate superior discriminative ability with AUC values of 95.1% and 95.3% respectively

Outcome and MACE definition



152 patients (11.2%) experienced MACE during the 1-year follow-up period after PCI

Feature Importance



SHAP analysis revealed the most influential predictors of MACE

Introduction

Cardiovascular diseases, specifically ischemic heart diseases (IHD), are the leading cause of mortality and morbidity across the world [1, 2]. The GBD 2019 study reported 197.2 million prevalent cases and 9.1 million deaths from IHD globally [3]. Across the widespread spectrum of IHD, patients with ST-elevation myocardial infarction (STEMI) have the worst prognosis [4].

The incidence of acute coronary syndromes increases with age [5, 6]. In the United States, over 60% of AMI (acute myocardial infarction) cases occur in individuals aged 65 years or older, and more than 60% of MI-related deaths occur in those aged 75 years or older [7]. Elderly patients often present with multiple comorbidities, including chronic kidney disease, diabetes, and hypertension, complicating their diagnosis and treatment [8]. These factors, along with reduced ejection fraction (EF), make them more susceptible to major adverse cardiovascular events (MACE) following IHD [6].

According to the most recent statements from the American Heart Association (AHA) guidelines and the European Society of Cardiology (ESC), older STEMI patients require a more specialized and personalized approach that considers frailty, cognitive function, and potential polypharmacy while balancing the risks of

bleeding and ischemic events [5, 9]. Despite these guidelines, outcomes in geriatrics remain suboptimal since their particular risk profiles are not fully understood [10]. A key reason is that their complex risk profiles are not adequately captured by existing prognostic tools.

Accurate and generalized risk stratification tools are crucial for this high-risk population. The prognosis in AMI is often evaluated using risk stratification models, such as the Thrombolysis in Myocardial Infarction (TIMI) risk score and the Global Registry of Acute Coronary Events (GRACE) score [11]. These traditional models typically rely on a limited set of predefined clinical variables and assume linear associations between predictors and outcomes [12, 13]. Additionally, they were developed based on studies primarily conducted on younger populations and may not fully capture the complexity of age-related physiological changes, frailty, and multiple comorbidities frequently seen in geriatric patients [14]. As a result, their predictive performance tends to decline in older adults, whose risk profiles are more heterogeneous and less well represented in the original derivation datasets [12, 13].

In contrast, machine learning (ML) algorithms are able to analyze large, high-dimensional clinical datasets. Moreover, unlike traditional models, they uncover

complex, nonlinear interactions among variables [15, 16]. Recent evidence highlights that incorporating a broader range of clinical, laboratory, and demographic features through ML approaches enables the development of more individualized and accurate risk prediction tools [17]. This is particularly relevant in elderly populations undergoing percutaneous coronary intervention (PCI), where age-related factors, comorbid conditions, and physiological variability limit the utility of traditional risk scoring systems [18, 19]. Consequently, prognostic prediction models developed using various ML algorithms have recently gained popularity for improving risk stratification for clinical decision-making [20–24].

Several recent studies have explored the application of ML in cardiovascular risk prediction. For example, Kwon et al. developed a deep-learning-based risk stratification model for the mortality of patients with AMI (DAMI) using data from 22,875 AMI patients and demonstrated significantly higher predictive performance than traditional scores such as GRACE and TIMI (e.g., area under the receiver operating characteristic curve (AUC) = 0.905 vs. 0.851 and 0.781, respectively) [25]. However, this and similar studies did not specifically address elderly PCI patients, who are the focus of our study. Most of these studies either focused on general populations rather than the elderly, lacked a comprehensive comparison of diverse ML algorithms, or did not perform interpretability analysis, which is crucial for clinical adoption.

To address these gaps, we developed and compared multiple ML algorithms, including logistic regression (LR), random forest (RF), gradient boosting (XGB), neural networks (NN), k-nearest neighbors (KNN), support vector machines (SVM), decision trees (DT), and naïve Bayes (NB), to predict 1-year MACE in elderly STEMI patients undergoing PCI. Unlike earlier studies, our work integrates domain-informed feature engineering, addresses data imbalance using the Synthetic Minority Oversampling Technique (SMOTE), and applies Shapley Additive Explanations (SHAP) analysis to enhance transparency and interpretability. By systematically benchmarking diverse models in a high-risk geriatric cohort, this study provides a practical and clinically relevant framework for individualized risk stratification.

Methods

Study design and population

This study employed a retrospective cohort design to predict the occurrence of 1-year MACE following PCI in elderly patients using ML models. This study utilized data from the Tehran Heart Center on 13,682 patients who underwent PCI from 2015 to 2021. From this population, we identified 5,910 patients aged 65 years or older. After applying the inclusion criteria, over one year of follow-up, 3,920 patients remained. The final study cohort

consisted of 1,358 patients for whom complete health records, laboratory tests, and echocardiography findings were available (Supplementary Figure 1).

Dataset description

The dataset comprised 1,358 patients aged 65 years or older who underwent PCI between 2015 and 2021. Patients had a mean age of 74.1 ± 6.7 years, and 31.8% of them were females. The average BMI was 27.4 ± 4.5 kg/m², and the mean Waist circumference was 99.2 ± 10.8 cm. The mean pre-PCI ejection fraction (EF) of the patients was $40.4 \pm 8.3\%$ and they had a door-to-device distance of 79.1 ± 66.2 minutes. Traditional cardiovascular risk factors were highly prevalent, with 61.1% hypertension, 48.5% diabetes mellitus, and 51.6% hyperlipidemia. In addition, 33.7% were smokers, 12.2% had a family history of CAD, 8.9% had prior CABG, and 12.8% had a history of PCI.

Laboratory findings showed a mean total cholesterol of 150.9 ± 37.7 mg/dL, triglycerides 124.2 ± 70.1 mg/dL, LDL 96.1 ± 32.9 mg/dL, HDL 39.2 ± 9.7 mg/dL, and LDL/HDL ratio 2.56 ± 0.98 . The mean fasting blood sugar (FBS) was 141.85 ± 65.39 mg/dL, hemoglobin 14.4 ± 1.9 g/dL, and creatinine 1.0 ± 0.4 mg/dL.

During the 1-year follow-up, 152 patients (11.2%) experienced MACE, including cardiovascular death, myocardial infarction, stroke, or repeat revascularization. Patients with MACE were generally older (76.5 vs. 73.9 years), had lower BMI and EF, higher creatinine and FBS, and worse lipid and hemoglobin profiles compared to those without MACE (all $p < 0.05$). These differences highlight the heterogeneity and complexity of the study population, emphasizing the need for robust predictive modeling approaches. Addressing the class imbalance was a critical step in model development, for which the SMOTE was applied to augment the minority class and improve predictive performance.

To handle these missing values (Less than 5%), we employed a median-mean imputation strategy, where missing entries were replaced with the median for numerical variables with skewed distributions and the mean for those with normal distributions.

Primary outcome and MACE definition

MACE was defined as a composite outcome frequently used in clinical trials and research to assess the risk or incidence of major cardiovascular events. For this study, MACE specifically included cardiovascular death, myocardial infarction, stroke, and revascularization procedures.

Follow-up and data collection

After performing the PCI, patients were closely monitored in the hospital for intra-hospital mortality and

morbidity. In addition, they were followed up for 30 days post-procedure and again at the 1-year mark to track long-term outcomes. Patients were prospectively followed up via phone calls, ensuring ongoing communication and monitoring of their health status after the procedure. These follow-up calls aimed to assess the progress of recovery, identify any complications or concerns, and gather additional clinical insights that may not have been captured during the initial medical visit. The follow-up process also allowed for the documentation of any new symptoms, medication changes, or treatment interventions, providing a more comprehensive understanding of the patients' post-procedural health.

Clinical, demographic, and laboratory data were extracted from electronic medical records. These data included pre-PCI EF, age, BMI, lipid profiles (e.g., LDL, HDL, cholesterol, and triglycerides), FBS, hemoglobin, creatinine levels, as well as relevant medical history, such as hypertension, diabetes, and hyperlipidemia.

Univariate analysis

Univariate analysis was conducted to examine the relationship between individual features and the outcomes of interest. Various clinical and demographic variables were tested for their significance in distinguishing between the non-MACE and MACE groups.

Statistical analysis

Statistical analysis was performed to assess the significance of differences between groups. Continuous variables were assessed using the t-test for normally distributed data and the Mann-Whitney U test for non-normally distributed data. Categorical variables were analyzed using the Chi-square test to determine associations between the variables and the outcomes of interest. In cases where the expected frequency was small (less than 5), the Fisher's exact test was applied to ensure accurate results. A p-value of less than 0.05 was considered statistically significant for all tests.

Data preprocessing for ML

1. **Outlier removal:** Before starting the analysis, outliers were removed through expert opinion or a double check.
2. **Feature selection:** Features showing significant differences between MACE and non-MACE groups were identified through univariate analysis. These factors were selected based on their significant association with the outcomes of interest and also their clinical relevance to the outcome.
3. **Removal of highly correlated parameters:** To enhance model efficiency and avoid redundancy, highly correlated variables were carefully removed,

and in cases of strong correlation, variables were combined, prioritizing those that provide the most comprehensive representation of the underlying factors. We used Chi-square tests for categorical variables and Pearson/Spearman correlation coefficients for continuous variables to assess bivariate relationships. This approach ensured that only the most relevant features were retained for further analysis, improving the predictive accuracy of the model.

4. **Normalization:** Continuous features were standardized using standard scaling (z-score normalization), ensuring a mean of 0 and a standard deviation of 1.
5. **Data Imbalance Handling:** To address the issue of data imbalance, the SMOTE was applied to augment the minority class (MACE) and improve model performance. SMOTE works by generating synthetic samples for the minority class rather than simply duplicating existing instances, which can help reduce overfitting and bias in the model (Eq. 1).

$$x_{new} = x_i + \lambda(x_i^* - x_i) \quad (1)$$

x_i is the selected sample from the minority class, x_i^* is one of its k-nearest neighbors,

λ is a random number between 0 and 1, which determines the position of the new synthetic point along the line segment between x_i and x_i^* .

The mechanism behind SMOTE involves selecting a data point from the minority class and identifying its k-nearest neighbors within the same class. A synthetic sample is then created by choosing a random point along the line segment connecting the selected data point and one of its neighbors. By generating synthetic data points, SMOTE helps create a more balanced dataset, ensuring that the model learns to classify both the majority and minority classes more effectively.

Train/Test splitting with K-fold cross-validation

To ensure robust model evaluation and mitigate the risk of overfitting, we employed K-fold cross-validation with $K=5$. In this approach, the dataset was randomly partitioned into five equal subsets (folds). For each iteration, four folds were used to train the model, while the remaining fold was used for validation. This process was repeated five times, with each fold serving as the validation set exactly once. The performance metrics from each iteration were averaged to obtain a more generalized estimate of the model's accuracy, precision, recall, F1 score, and AUC. This method allowed us to assess the model's stability and generalizability across different subsets of the data and ensured that the evaluation was not biased by a particular train-test split.

To ensure a robust and unbiased evaluation of our models, feature selection was performed within each cross-validation fold to prevent information leakage. Specifically, for each fold in the 5-fold cross-validation process, feature selection was applied only to the training data within that fold, and the selected features were then used to train the model and evaluate it on the corresponding test fold. This approach ensures that no information from the test set influenced the feature selection process, maintaining the integrity of the cross-validation results.

ML models

The study employed a diverse range of ML models to analyze the data and predict outcomes. These models included:

1. **XGradient Boosting (XGB)**: A powerful ensemble learning algorithm known for its exceptional performance in classification tasks. It is used with specific settings to disable label encoding (`use_label_encoder = False`) and optimize evaluation using the log loss metric (`eval_metric = 'logloss'`). Default hyperparameters include a learning rate (`eta = 0.3`), maximum tree depth (`max_depth = 6`), and L2 regularization (`lambda = 1`).
2. **Random Forest (RF)**: An ensemble method based on decision trees. It evaluates the quality of splits using the entropy criterion (`criterion = 'entropy'`) to improve classification accuracy. Default settings include 100 trees (`n_estimators = 100`), Gini impurity as the default criterion, and bootstrapping enabled (`bootstrap = True`).
3. **Logistic Regression (LR)**: A statistical model designed for binary classification. It applies L2 regularization (`penalty = 'l2'`) and uses a solver (`solver = 'lbfgs'`) designed for efficiency, with a high maximum iteration limit (`max_iter = 1000`) to ensure proper model training. The default inverse regularization strength is `C = 1.0`.
4. **Neural Network (NN)**: A multilayer perceptron model that leverages a network of interconnected layers for classification tasks. It is trained over a large number of iterations (`max_iter = 1000`) to ensure adequate learning. Default parameters include a single hidden layer with 100 neurons (`hidden_layer_sizes = (100)`), ReLU activation (`activation = 'relu'`), and the Adam optimizer (`solver = 'adam'`).
5. **Support Vector Machine (SVM)**: A classifier using a linear kernel (`kernel = 'linear'`) for decision boundaries. It is configured to provide probability estimates (`probability = True`) for its predictions. The default regularization parameter is `C = 1.0`, and it uses hinge loss for optimization.
6. **Naive Bayes (NB)**: A probabilistic classification model based on Bayes' theorem, particularly effective for continuous data using the Gaussian distribution (GaussianNB). Default parameters include a prior probability of classes based on data (`priors = None`) and additive smoothing (`var_smoothing = 1e-9`).
7. **K-Nearest Neighbors (KNN)**: A classification algorithm that identifies the class of a data point by analyzing the nearest neighbors within a defined neighborhood size (`n_neighbors = 5`). Default settings include uniform weighting (`weights = 'uniform'`), Euclidean distance metric (`p = 2`), and the brute-force algorithm for small datasets (`algorithm = 'auto'`).
8. **Decision Tree (DT)**: A non-parametric supervised learning model that splits data into branches for decision-making, aiming to classify effectively without assumptions about data distribution. Default parameters include Gini impurity (`criterion = 'gini'`), unlimited maximum depth (`max_depth = None`), and a minimum of two samples required to split a node (`min_samples_split = 2`).

Model evaluation

The models were evaluated using a range of performance metrics to assess their predictive accuracy. Key metrics included accuracy, precision (positive predictive value), sensitivity (recall), specificity, and AUC. AUC reflects the model's ability to discriminate between patients with and without MACE across various decision thresholds, with higher AUC values indicating better overall performance.

Accuracy provides an overall measure of correct predictions, while precision focuses on the reliability of positive predictions. Sensitivity and specificity highlight the model's ability to detect true positives and true negatives, respectively, ensuring minimal false positives and negatives. The AUC value assesses the model's ability to distinguish between classes, with higher values indicating better classification performance. The optimal cut-off point was selected based on iterative threshold testing to balance sensitivity and specificity.

The F1 score is particularly important in imbalanced datasets, as it balances precision and recall. It is calculated using the following formula (Eq. 2)

$$F1 = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \quad (2)$$

Hyperparameter optimization

Hyperparameters were tuned via grid search using cross-validation. We empirically tested multiple combinations for each algorithm and selected those that yielded the best validation performance (Supplementary Table 1).

Explainability analysis

SHAP is a widely accepted tool for enhancing the interpretability of ML models, particularly in medical applications where clinical decisions depend on trust and transparency. It helps clinicians understand which features most influence model predictions, making complex models more accessible to non-technical users. SHAP Feature Importance Plot is used to visualize the ranking of features based on their mean absolute SHAP values. By ranking features according to their contribution, this plot allowed us to identify which variables had the most substantial impact on predicting MACE.

In addition to the feature importance plot, the SHAP Summary Plot was used to visualize the relationship between feature values and their influence on the model's predictions. This plot displayed how each feature's values (e.g., high or low) affected the model's output, with higher SHAP values indicating a stronger influence on the risk of MACE. It provided an intuitive way to assess the direction and magnitude of the effects of different features, further enhancing the model's transparency and helping clinicians understand which factors most significantly impacted patient outcomes.

Software and tools

All analyses were performed in Python. Libraries included scikit-learn for ML pipelines, XGB for gradient boosting, imbalanced-learn for SMOTE, and SHAP for model explainability. Data manipulation and visualization were conducted using Pandas, NumPy, Matplotlib, and Seaborn. Univariate analysis was performed using SPSS version 27 to examine the relationship between individual features and the outcomes of interest.

Ethical considerations

This study was conducted following ethical guidelines and approved by the relevant institutional review board (IRB) of Tehran Heart Center (THC) with an Ethical approval number of IR.TUMS.TH.CREC.1399.045. Given its retrospective nature and utilizing de-identified data from THCs electronic registry, informed consent was waived for the analysis of anonymized patient data. All patient information used in this study was de-identified to protect privacy and confidentiality. The data were extracted from electronic medical records, which were maintained in compliance with the Health Insurance Portability and Accountability Act (HIPAA) or equivalent privacy laws to ensure the security of sensitive patient information.

Additionally, the study adhered to ethical principles outlined in the Declaration of Helsinki, ensuring that all research involving human participants was conducted with respect for their rights, safety, and well-being. The primary goal of the study was to advance clinical

knowledge and improve patient outcomes by identifying risk factors for major adverse cardiac events in elderly patients following PCI. All analyses and reporting were performed with the utmost consideration for maintaining patient anonymity and avoiding any harm or discomfort to the participants.

The graphical abstract showed a summary of methods and results (Fig. 1).

Results

Demographic characteristics of the study population

In this study, we enrolled 1,358 elderly participants, all of whom underwent PCI and were followed for one year. The mean age was 74.1 ± 6.7 years, and 433 (31.8%) were female. The mean BMI was 27.4 ± 4.5 . The mean pre-PCI EF was $40.4 \pm 8.3\%$. The average door-to-device time was 79.1 ± 66.2 minutes, and the prevalence of hyperlipidemia, diabetes, and hypertension were 51.6%, 48.5%, and 61.1%, respectively, while 459 (33.7%) participants were smokers. Among these patients, 152 (11.2%) patients experienced MACE during the 1-year follow-up after PCI.

Univariate analysis

The univariate analysis shows that patients in the MACE group had significantly lower pre-PCI EF (37.1% vs. 40.9%, $p < 0.01$), higher age (76.5 vs. 73.9 years, $p < 0.01$), lower BMI, and waist circumference compared to the non-MACE group. Additionally, total cholesterol (141.2 vs. 152.2 mg/dL, $p < 0.01$) and LDL levels (87.3 vs. 97.3 mg/dL, $p < 0.01$) differed across groups. In addition FBS (166 vs. 138.8 mg/dL, $p < 0.01$) were significantly higher in MACE group. Lower hemoglobin levels (13.6 vs. 14.4 g/dL, $p < 0.01$) and higher creatinine levels (1.2 vs. 1.0 mg/dL, $p < 0.01$) were also observed in the MACE group, indicating possible associations with poorer cardiac outcomes (Supplementary Figure 2).

Other clinical and demographic factors, such as gender, hypertension, smoking, opium use, family history of coronary artery disease (CAD), and previous interventions (CABG or PCI), did not significantly differ between the groups. Hyperlipidemia and diabetes were more prevalent in the non-MACE group (52.57% vs. 44.74%, $p = 0.04$) and (57.24% vs. 47.43%, $p = 0.01$), respectively. Some variables, such as triglycerides, HDL, and door-to-device time, did not show significant differences, suggesting they may not independently contribute to distinguishing between the two groups. Overall, this data highlights key metabolic, demographic, and clinical markers associated with the occurrence of MACE events (Table 1).

Feature selection and hyperparameter optimization

After conducting univariate analysis, we identified features that significantly differed between the non-MACE

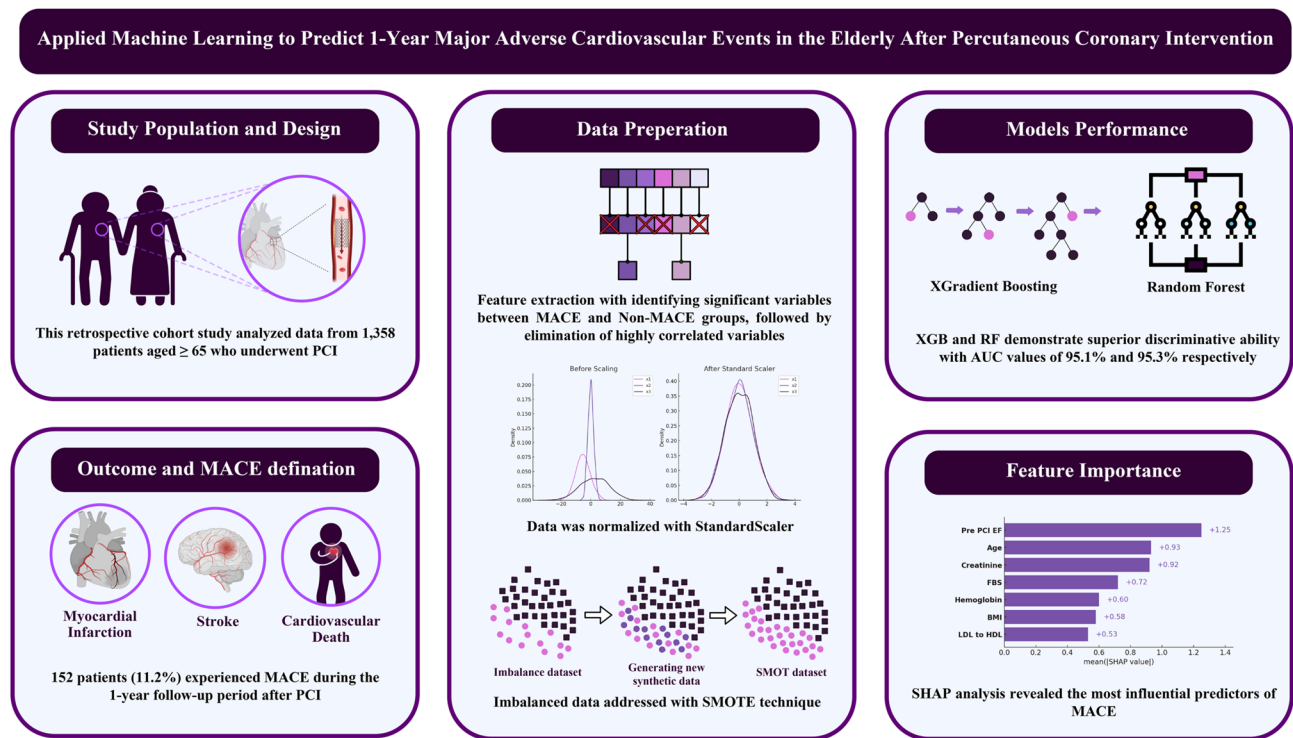


Fig. 1 A summary of methods and results

and MACE groups. These features included pre-PCI EF, age, BMI, waist circumference, total cholesterol, LDL levels, LDL/HDL ratio, FBS, hemoglobin, and creatinine, as well as the prevalence of hyperlipidemia and diabetes.

Since BMI and waist circumference were highly correlated, we selected BMI as it reflects both weight and height combined. Instead of considering LDL, HDL, and total cholesterol individually, we used the LDL/HDL ratio, which serves as a comprehensive marker representing lipid profile characteristics. Hyperlipidemia and diabetes were strongly correlated with FBS and lipid profiles. To avoid redundancy, we excluded these conditions and instead prioritized representative markers of lipid and glucose profiles, which also provide insights into treatment adherence (Fig. 2).

Evaluation metrics of the ML models

As mentioned in Table 2, Among the evaluated models, RF achieved the highest overall performance with an AUC of 0.953 (95% CI: 0.943–0.964), combining excellent accuracy (0.909), precision (0.923), specificity (0.967), and negative predictive value (NPV: 0.903), alongside strong sensitivity (0.793) and F1 score (0.853). XGB followed closely with an AUC of 0.941, showing a balanced trade-off between precision (0.849), recall (0.775), and specificity (0.935), though with slightly lower accuracy (0.84). Traditional Logistic Regression, Neural Networks, and SVM performed moderately (AUCs ~ 0.81),

but showed weaker recall, particularly SVM (0.522) and Logistic Regression (0.529), reducing their F1 scores. Naive Bayes and KNN offered slightly better recall (0.705 and 0.721, respectively), but had lower precision and accuracy. The Decision Tree, while having a relatively lower AUC (0.792), demonstrated solid recall (0.755) and specificity (0.830), with a competitive accuracy of 0.805. Overall, ensemble methods (RF and XGB) outperformed others across most metrics, highlighting their robustness in this prediction task (Table 2, Fig. 3, and Supplementary Figure 3 and Supplementary Table 2). Overall, XGB and RF emerge as the most robust models, balancing strong performance across all metrics.

Model selection

Although both RF and XGB emerged as the top-performing models with similar discrimination, we chose XGB as the primary model for SHAP-based explainability due to its algorithmic suitability for gradient-based feature impact analysis.

Interpreting model insights through SHAP analysis

Figure 4 demonstrated SHAP analysis to interpret the importance and impact of features in a predictive model. Figure 4(A) ranks features by their mean absolute SHAP values, reflecting their overall contribution to the model. The most influential feature is pre-PCI EF, with the highest mean SHAP value (+1.25), followed by age (+0.93),

Table 1 Univariate analysis of demographic characteristics, risk factors, clinical features, and laboratory parameters of the study population

	Total population N = 1358	Non-MACE N = 1206	MACE N = 152	P value
Demographic characteristics				
Age, year, Mean $\pm$ SD	74.1 $\pm$ 6.7	73.9 $\pm$ 6.6	76.5 $\pm$ 7.8	< 0.01*
Gender, female, N (%)	433 (31.8%)	377 (31.2%)	56 (36.8%)	0.16
BMI, Kg/m ² , Mean $\pm$ SD	27.4 $\pm$ 4.5	27.5 $\pm$ 4.5	26.6 $\pm$ 4.7	0.02*
Waist circumference, cm, Mean $\pm$ SD	99.2 $\pm$ 10.8	99.6 $\pm$ 10.7	96.8 $\pm$ 11.8	< 0.01*
Risk Factors				
Hyperlipidemia, N (%)	702 (51.6%)	634 (52.5%)	68 (44.7%)	0.04*
Diabetes, N (%)	659 (48.5%)	572 (47.4%)	87 (57.2%)	0.01*
Hypertension, N (%)	831 (61.1%)	741 (61.4%)	90 (59.2%)	0.32
Family history of CAD, N (%)	167 (12.2%)	151 (12.5%)	16 (10.57%)	0.28
Smoking, N (%)	459 (33.7%)	403 (33.45%)	56 (36.8%)	0.41
Previous CABG, N (%)	122 (8.9%)	109 (9.0%)	13 (8.5%)	0.49
Previous PCI, N (%)	175 (12.8%)	153 (12.6%)	22 (14.4%)	0.52
Clinical characteristics				
Pre PCI EF, %, Mean $\pm$ SD	40.4 $\pm$ 8.3	40.9 $\pm$ 8.0	37.1 $\pm$ 10.2	< 0.01*
Door to Device Distance, min, Mean $\pm$ SD	79.1 $\pm$ 66.2	79.0 $\pm$ 65.9	80.5 $\pm$ 69.3	0.78
Laboratory findings				
Total Cholesterol, mg/dL, Mean $\pm$ SD	150.9 $\pm$ 37.7	152.2 $\pm$ 37.6	141.2 $\pm$ 37.5	< 0.01*
Triglyceride, mg/dL, Mean $\pm$ SD	124.2 $\pm$ 70.1	124.6 $\pm$ 69.5	121.3 $\pm$ 75	0.57
LDL, mg/dL, Mean $\pm$ SD	96.1 $\pm$ 32.9	97.3 $\pm$ 33.1	87.3 $\pm$ 30.4	< 0.01*
HDL, mg/dL, Mean $\pm$ SD	39.2 $\pm$ 9.7	39 $\pm$ 10	38 $\pm$ 9	0.20
LDL to HDL ratio, Mean $\pm$ SD	2.56 $\pm$ 0.98	2.6 $\pm$ 1.0	2.4 $\pm$ 0.9	0.01*
FBS, mg/dL, Mean $\pm$ SD	141.85 $\pm$ 65.39	138.8 $\pm$ 61	166 $\pm$ 89.7	< 0.01*
Hemoglobin, mg/dL, Mean $\pm$ SD	14.4 $\pm$ 1.9	14.4 $\pm$ 1.8	13.6 $\pm$ 2.2	< 0.01*
Creatinine, mg/dL, Mean $\pm$ SD	1.0 $\pm$ 0.4	1.0 $\pm$ 0.4	1.2 $\pm$ 0.5	< 0.01*

BMI: Body Mass Index, CAD: Coronary Artery Disease, EF: Ejection Fraction, FBS: Fasting Blood Sugar, HDL: High-Density Lipoprotein, LDL: Low-Density Lipoprotein, MACE: Major Adverse Cardiac Events, PCI: Percutaneous Coronary Intervention, SD: Standard Deviation

creatinine (+0.92), and other variables like FBS, hemoglobin, BMI, and LDL/HDL ratio. These results suggest that the model relies heavily on these factors to make predictions.

Figure 4(B) provides a detailed view of how each feature's values impact the model's output. As demonstrated, lower values of pre-PCI EF and age (red) are associated with an increased risk of MACE among patients. Similarly, elevated levels of creatinine and FBS are linked to a higher risk of MACE, reflecting the critical roles of renal function and diabetes in cardiovascular outcomes.

Additionally, a decrease in the LDL/HDL ratio corresponds to an increased risk of MACE, highlighting its significance in lipid metabolism. Likewise, as BMI decreases, the risk of MACE tends to rise, suggesting the potential adverse effects of low body weight on patient outcomes. These findings underscore the complex interactions between clinical factors and cardiovascular risk.

Discussion

Despite the high prevalence of geriatrics across various populations, they are always underrepresented in clinical trials, mostly due to multimorbidity, polypharmacy, and

safety issues [26]. One approach to address this issue is the implementation of AI, specifically ML algorithms in electronic health records [27]. This study highlights the significant clinical factors associated with MACE risk in elderly patients post-PCI. ML models, particularly XGB, demonstrated strong performance with an AUC of 95.1%, accuracy of 88.4%, precision (PPV) of 84.9%, NPV of 89.8%, sensitivity of 77.5%, specificity of 93.5%, and an F1 score of 81.08%. Key features such as pre-PCI EF, age, creatinine levels, FBS, BMI, and lipid profile contributed most to the prediction of MACE. These findings emphasize the importance of monitoring metabolic and clinical markers for better risk stratification and managing elderly PCI patients.

Several previous studies have explored ML-based predictive models for MACE risk assessment (Table 3). For example, Zhang et al. (2024) used ML models to predict MACE after the first PCI in patients with acute MI, reporting that XGB achieved an AUC of 83.6% [22]. Moreover, Lee et al. (2022) used only 2 ML models, LR and RF, with AUCs of 75.3% and 74.6%, respectively, for predicting 1-year survival in elderly patients with MI [33]. Our study showed even higher AUC values (95.1%

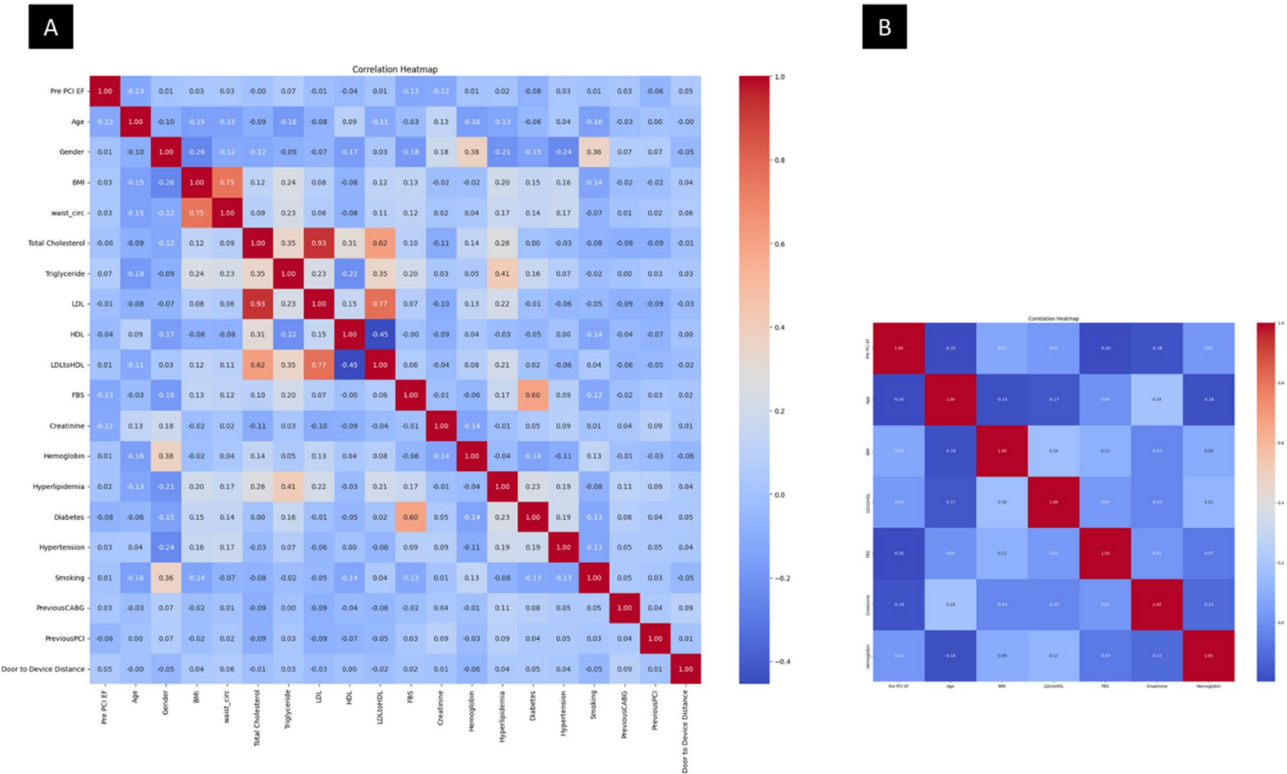


Fig. 2 (A) heatmap showing the correlation between parameters. (B) heatmap after removing highly correlated parameters, highlighting candidates for ML

Table 2 Performance metrics comparison of various ML models

Models	AUC	Confidence Interval	Accuracy	Recall (Sensitivity)	Precision (PPV)	NPV	Specificity	F1 Score
Random Forest	0.953	0.943 - 0.964	0.909	0.793	0.923	0.903	0.967	0.853
XGB	0.941	0.930- 0.950	0.84	0.775	0.849	0.898	0.935	0.81
Logistic Regression	0.81	0.787 - 0.832	0.771	0.529	0.71	0.791	0.892	0.606
Neural Net	0.808	0.786 - 0.829	0.758	0.592	0.651	0.805	0.842	0.62
SVM	0.812	0.789 - 0.834	0.778	0.522	0.736	0.791	0.906	0.611
Naive Bayes	0.778	0.753 - 0.800	0.722	0.705	0.567	0.832	0.731	0.629
KNN	0.816	0.795 - 0.834	0.748	0.721	0.602	0.845	0.761	0.656
Decision Tree	0.792	0.772 - 0.813	0.805	0.755	0.689	0.871	0.83	0.721

AUC: Area Under the Curve, KNN: K-Nearest Neighbors, NPV: Negative Predictive Value, PPV: Positive Predictive Value (also known as Precision), SVM: Support Vector Machine

for XGB and 95.3% for RF), possibly due to the inclusion of additional key predictors such as LDL/HDL ratio and FBS. Similarly, Wu et al. (2023) developed several ML-based models to predict major adverse cardiovascular and cerebrovascular events in older patients undergoing non-cardiac surgery. Of these, the best performance exhibited was from the XGB model, which had an AUC of 90.3% [23]. Likewise, Liu et al. (2024) used the SMOTE

technique to address positive and negative imbalance issues and achieved an AUC of 85% with the RF model for predicting mortality in hypertensive heart failure patients [34]. The superior performance of our models highlights the effectiveness of ML in cardiovascular risk stratification, specifically when addressing data imbalance issues and utilizing domain-specific feature selection techniques. Additionally, a recent study by Yavuz et al. 2025

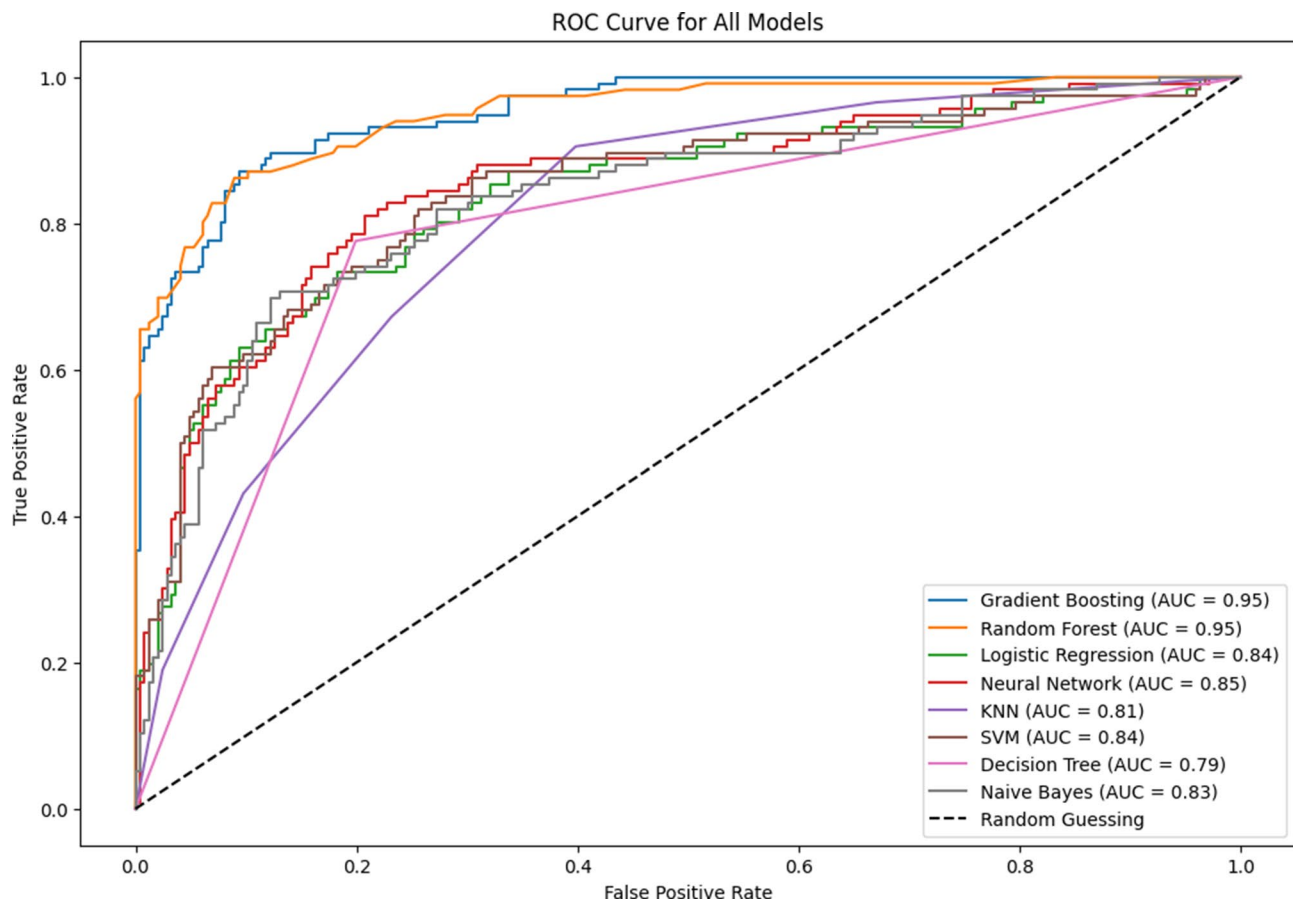


Fig. 3 AUC for various machine learning models, with XGB and RF emerging as the most robust models, demonstrating strong performance across all metrics

demonstrated the potential of ML models in predicting contrast-induced nephropathy (CIN) in acute coronary syndrome patients by integrating shock indices into a Gradient Boosting Machine-based model (CIN-Predict 5), achieving superior discrimination (AUC=0.87) compared to the Mehran Risk Score [35]. This further highlights the value of ML approaches in addressing procedure-related complications and supports their integration into clinical workflows for comprehensive risk stratification in PCI populations.

Lipid profile and its association with risk for MACE remains controversial [36, 37]. Though hyperlipidemia is an established risk factor for cardiovascular disease [38], in our study, the levels of LDL were lower among the MACE group, possibly indicating poor nutritional status in elderly patients [39–41]. In contrast to God et al. 's findings, which indicated no correlation between MACE and the ratio of LDL/HDL in patients with MI (not only geriatrics) [42], our results suggest that the LDL/HDL ratio may be more informative than absolute LDL levels for risk stratification in this population.

The LDL/HDL ratio is a significant predictor of outcomes in patients with cardiovascular conditions,

particularly those undergoing percutaneous coronary intervention (PCI) or experiencing MI. In STEMI patients undergoing PCI, a higher LDL/HDL ratio (≥ 2.59) is an independent risk factor for long-term coronary revascularization, with a significantly higher revascularization rate compared to lower and medium ratio groups [43]. Additional research indicates a complex relationship between this ratio and clinical outcomes. For patients undergoing PCI, a V-shaped association has been noted between HDL-C levels and mortality risk, with HDL-C levels exceeding 60mg/dL associated with a sharply increased mortality risk [44]. Also, a U-shaped association has been observed in elderly hypertensive patients, where both lower and higher LDL/HDL ratios (< 1.67 and ≥ 2.10) are linked to increased all-cause mortality [45]. These findings highlight the importance of monitoring and managing the LDL/HDL ratio to optimize long-term outcomes in patients with PCI and MI.

Consistent with the “obesity paradox” in cardiovascular diseases, which declares worse prognosis in normal patients compared to mildly obese patients [46], our model revealed that lower BMI was associated with a higher risk of MACE in elderly patients post-PCI. a

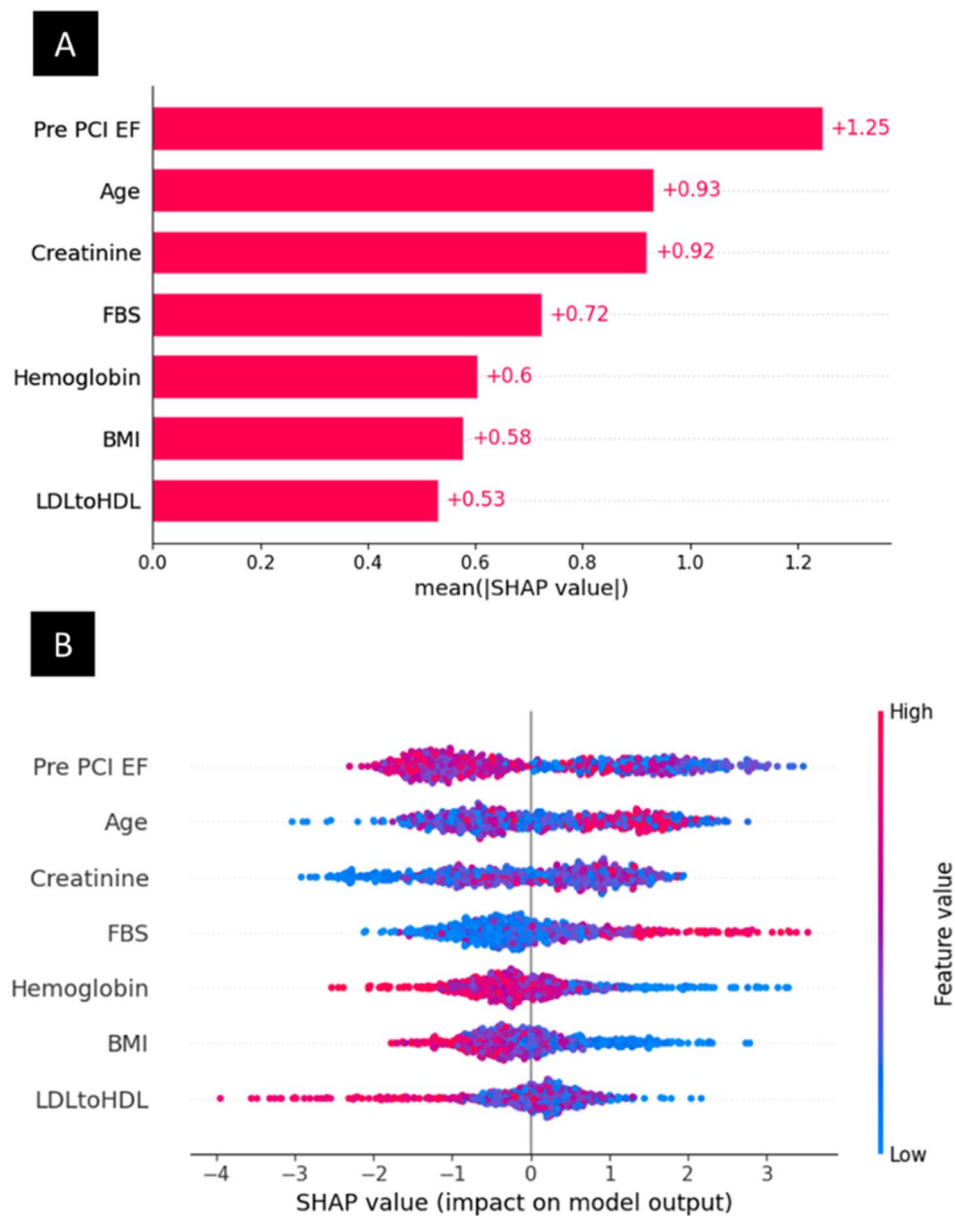


Fig. 4 SHAP analysis of feature importance and impact on MACE risk. **(A)** features were ranked by their contribution to the model, with pre-PCI EF, age, and creatinine identified as the most influential predictors. **(B)** SHAP summary plot showing how high or low feature values (red = high, blue = low) impact MACE risk. High pre-PCI EF, age, creatinine, and FBS increase risk, while lower BMI and LDL/HDL ratio are also associated with higher MACE risk

registry-based study in India demonstrated increased incidence of MACE in patients with low BMI [47]. However, He et al. declined the aforementioned paradox in patients older than 75 years old and degraded the value of BMI in predicting MACE in geriatrics [48]. This finding underscores the importance of considering body weight and nutritional status in the management of these patients, as low BMI may reflect frailty or malnutrition, contributing to adverse outcomes.

In our study, traditional cardiovascular risk factors including hypertension, smoking, and previous PCI did not differ significantly between MACE and non-MACE

groups in elderly patients undergoing primary PCI. This finding aligns with the well-documented concept of “reverse epidemiology” or “risk factor paradox” in elderly populations, a phenomenon describing how traditional cardiovascular risk factors that predict poor outcomes in younger populations may show opposite or neutral associations in older adults [49]. Furthermore, elderly patients who survive to undergo procedures like PCI represent a selected population that has already survived the early lethal effects of cardiovascular risk factors, creating a survival bias that attenuates the apparent predictive power of these traditional risk factors [50]. Additionally,

Table 3 Literature review of the studies using machine learning for the prediction of major adverse cardiovascular events after percutaneous coronary intervention

Author (Year)	Population	ML Models	Variables Included in Model	Performance (Auc)	Smote	Limitations
Zhang et al. (2024) [22]	General	ANN, KNN, SVM, RF, LR	Demographics, Comorbidities, Laboratory Test, Preoperative PCI Characteristics, PCI Procedure Characteristics	ANN: 0.8049; RF: 0.7267; KNN: 0.798; SVM: 0.772; LR: 0.7177	No	Survival bias, missing data Handling, Limited External Validation
Niimi et al. (2022) [28]	General	XGB	AKI Model Variables, Bleeding Model Variables, In-hospital Mortality Model Variables	XGB: 0.84	No	Imbalanced dataset, Missing data, Limited population, Potential selection bias, No use of balancing techniques, Limited to XGBoost algorithm only, Temporal validation instead of external validation, Limited to in-hospital outcomes, No long-term follow-up data
Wang et al. (2022) [29]	General	DT, RF, LR, NB, SVM, XGBoost	Demografics, Clinical factors, Laboratory Test, Medications	XGB: 0.86 (Best)	Yes	Limited Data Size, Limited Source Population, Structured Data Limitation
Liu et al. (2021) [30]	General	SVM, DT, RF, Gradient boosting decision tree, NN, LR	Demographics, Comorbid Conditions, Electrophysiology, Coronary Angiography, PCI Procedure, Medications, Laboratory Test, Echocardiogram	RF: 0.71 (Best)	No	Single-center, no external validation, clinical impact not assessed, potential missing predictors
Juan-Salvadores et al. (2022) [31]	Young	SVM, RF, NB, MLP, LDA, LR	Demographics, Risk Factors and Medical History, Laboratory Test, Clinical Parameters	RF: 0.80 (1 year), 0.79 (long-term) (Best)	No	Small sample size, limited generalizability to older populations, Limited sample size, Restricted number of variables in database, Small number of MACE events
Kim et al. (2022) [32]	General	DNN, GBM, GLM	Demographics, Laboratory Test	DNN: 0.94–0.97 (various timepoints), GBM: 0.95–0.96 (various timepoints)	No	No external validation, Dataset limited to Korean patients only, Limited to 1-year follow-up period, Missing details about medication dosage and duration, Potential dataset bias

ANN: Artificial Neural Network, KNN: K-Nearest Neighbors, SVM: Support Vector Machine, RF: Random Forest, LR: Logistic Regression, PCI: Percutaneous Coronary Intervention, XGBoost: Extreme Gradient Boosting, AKI: Acute Kidney Injury, DT: Decision Tree, NB: Naive Bayes, NN: Neural Network, MLP: Multilayer Perceptron, LDA: Linear Discriminant Analysis, DNN: Deep Neural Network, GBM: Gradient Boosting Machine, GLM: Generalized Linear Model, MACE: Major Adverse Cardiovascular Events

in elderly populations with established cardiovascular disease, conditioning on survival or disease presence can create collider bias, which occurs when analyzing selected populations such as elderly PCI patients where the selection process itself is influenced by both the exposure (risk factors) and outcome (survival to procedure). These methodological considerations, combined with competing mortality risks and the emergence of frailty as a dominant predictor in geriatric populations, underscore the urgent need for developing novel predictive models and identifying age-specific biomarkers that can more accurately stratify risk in elderly patients.

Clinical Implications

We proposed practical and feasible algorithms that used basic variables for predicting MACE, mostly available across all types of healthcare settings. High-risk patients could be flagged for intensified post-discharge monitoring, early follow-up, or multidisciplinary management, while the model's transparency preserves clinician oversight. For patients flagged as high-risk, clinicians could

intensify optimization of modifiable metabolic and clinical factors (e.g., tighter glycemic control, renal function monitoring, and reconsideration of lipid/nutritional status), incorporate comprehensive frailty and nutritional assessments, and arrange closer post-discharge surveillance (earlier follow-ups, telemonitoring, or enrollment in cardiac rehabilitation). For real-world applications, the proposed ML models could be integrated into the electronic health record as a decision support tool to provide individualized 1-year MACE risk estimates at the time of PCI. Prior to deployment, external validation, calibration assessment, and prospective pilot testing would be necessary to ensure generalizability and safety. Continuous monitoring for model drift, subgroup performance evaluation to guard against bias, and a governance framework would be essential to sustain trust and effectiveness in clinical practice.

Limitations

It is important to note that our study has several limitations:

First, a major limitation was the intrinsic class imbalance, with only 11.2% of patients developing MACE. To mitigate this, we applied the SMOTE technique to generate synthetic minority class samples and improve model performance. Second, similar to most retrospective studies, our study is subject to survival bias in that patients with the poorest outcomes may have been excluded from long-term follow-up. As declared by Banack et al., survival bias could interfere with real results in geriatric studies [51]. In particular, older and more frail patients who may have experienced early mortality could have been underrepresented in our dataset. This underrepresentation might bias the model toward better outcomes and reduce the accuracy of predictions for the most vulnerable subgroups. Third, although previous studies have highlighted that optimal guideline-directed medical therapy has a significant impact on post-PCI outcomes, our study did not investigate the effect of pharmaceutical treatment on MACE incidence [52, 53]. In addition, we acknowledge the potential influence of unmeasured confounders such as medication adherence, frailty, and social determinants of health, which were not captured in our dataset. These factors could meaningfully affect patient outcomes, and we recommend that future prospective study designs account for these variables. Fourth, our study dataset lacks information on inflammatory markers or endothelial dysfunction markers, which prior studies have emphasized as important in cardiovascular disease and acute coronary syndrome, particularly in elderly patients and those with renal disease, for determining outcomes [54, 55]. Fifth, our study is a single-center, retrospective study with a relatively modest sample size, which limits the generalizability of our results. Moreover, we did not perform external validation, which further limits the applicability of our findings across different healthcare settings. Future studies should use multi-center datasets to validate these findings in wider populations and address previous issues. Sixth, while SHAP analysis provided valuable insights into the contribution of predictors such as pre-PCI EF, age, and creatinine, its interpretation is sensitive to the underlying dataset size and model complexity. SHAP values in smaller, single-center cohorts may overestimate feature importance, underscoring the need for external validation and larger datasets to confirm feature rankings and ensure generalizability. Finally, it is suggested that future research focus on incorporating explainable AI frameworks, such as LivXAI-Net, to improve the transparency of machine learning models and bolster real-time clinical decision support systems [56]. By leveraging these frameworks, researchers can enhance the interpretability of complex AI models, ensuring that clinicians understand the reasoning behind predictions.

Conclusion

Our study demonstrates that ML-driven models can accurately forecast adverse cardiovascular events in elderly post-PCI patients by leveraging key predictors—such as ejection fraction, age, renal function, glycemic control, and lipid ratios—to enhance risk stratification and guide personalized care. Embedding these algorithms into electronic health records (EHRs) as clinical decision support tools could enable real-time, point-of-care risk scores that alert clinicians to high-risk individuals and inform tailored treatment strategies, ranging from optimized pharmacotherapy and targeted monitoring to timely referrals for rehabilitation or telehealth follow-up.

Looking ahead, future research should focus on multicenter external validation, prospective clinical trials, and the development of deep learning architectures capable of integrating multimodal data (e.g., imaging, longitudinal laboratory trends) to improve predictive power. Additionally, studies should evaluate the feasibility, cost-effectiveness, and clinical impact of integrating these models into routine care workflows and monitor performance for model drift over time. Such efforts will help ensure safe, equitable, and sustainable use of AI-enhanced cardiovascular risk prediction in real-world clinical settings.

Medical and clinical abbreviations

CVD	Cardiovascular Disease
IHD	Ischemic Heart Disease
STEMI	ST-Elevation Myocardial Infarction
MI	Myocardial Infarction
MACE	Major Adverse Cardiac (or Cardiovascular) Events
PCI	Percutaneous Coronary Intervention
EF	Ejection Fraction
CAD	Coronary Artery Disease
CABG	Coronary Artery Bypass Grafting
AHA	American Heart Association
ESC	European Society of Cardiology
LDL	Low-Density Lipoprotein
HDL	High-Density Lipoprotein
FBS	Fasting Blood Sugar
BMI	Body Mass Index
NPV	Negative Predictive Value

Statistical and computational abbreviations

ML	Machine Learning
SHAP	Shapley Additive Explanations
SMOTE	Synthetic Minority Oversampling Technique
AUC	Area Under the (Receiver Operating Characteristic) Curve
F1 Score	Harmonic Mean of Precision and Recall
ROC	Receiver Operating Characteristic

Machine learning model abbreviations

XGB	XGradient Boosting
RF	Random Forest
LR	Logistic Regression
NN	Neural Network
KNN	K-Nearest Neighbors
SVM	Support Vector Machine
DT	Decision Tree
NB	Naive Bayes

Supplementary Information

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Supplementary Material 1

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Author contributions

A.G.J. contributed to data formal analysis and writing - original draft. A.N., Ali N., and H.P.S. contributed to data curation. A.G.J., A.N., Ali N., H.P.S., and A.A. contributed to writing - review & editing. A.A. contributed to visualization. Conceptualization was carried out by Y.J., M.M., K.H., and H.S.

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Data availability

The data supporting this study's findings are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

The research was conducted in accordance with the Declaration of Helsinki and received approval from the Ethics Committee by the relevant institutional review board (IRB) of Tehran Heart Center (THC), with an Ethical approval number of IR.TUMS.TH.CREC.1399.045. The Institutional Ethics Committee waived the requirement of informed consent because it was a retrospective study. All methods were performed in accordance with the relevant guidelines and regulations.

Consent for publication

Not applicable.

Competing interest

The authors declare no competing interests.

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